BEAUTY 360°

BUSINESS REQUIREMENTS DOCUMENT (BRD)

Beauty E-commerce Catalogue Intelligence | Version 1.0 | Portfolio Project

Project	Beauty 360° – Beauty Catalogue Intelligence	**Tools**	Python, PostgreSQL, Power BI
Data source	Sephora Product Catalogue	**Analysis type**	Catalogue snapshot analysis

1. Project Overview

BEAUTY 360° is an end-to-end analytics project that examines a beauty e-commerce product catalogue and turns it into business-focused insights. The workflow covers data profiling, cleaning, exploratory data analysis (EDA), feature engineering, SQL-based transformation and validation, and interactive Power BI reporting.

The project focuses on catalogue structure, brand and category performance, price positioning, product availability, and customer engagement indicators. It is designed as a practical portfolio case study for beauty and personal-care retailers.

2. Business Problem

Beauty retailers manage large catalogues across many brands, categories and price points. Without a consistent analytical view, teams may struggle to understand where catalogue breadth is concentrated, which brands attract customer interest, how prices differ across categories, and where listed products are unavailable.

This project creates a validated reporting dataset and a compact dashboard experience to help business users explore these questions. It provides descriptive evidence for catalogue and merchandising discussions; it does not claim to measure sales or profit.

3. Project Objectives

1. Prepare a reliable analysis-ready dataset through documented cleaning decisions.
2. Create useful analytical features, including price segments.
3. Use PostgreSQL to stage, transform and validate the reporting table.
4. Compare categories and brands using catalogue and engagement measures.
5. Build three Power BI analysis pages with clear KPIs, visuals and business explanations.
6. Translate findings into practical recommendations while stating data limitations.

4. Dataset Overview

Attribute	Project dataset
Product records	8,494
Distinct brands	304
Product categories	9
Data type	Product catalogue snapshot
Currency	USD

Interpretation: the dataset represents a catalogue snapshot, not a time-series or transaction-level dataset. Counts and metrics describe the records available in this dataset.

BUSINESS REQUIREMENTS & ANALYTICAL QUESTIONS

5. Stakeholders and Their Needs

Stakeholder	Information need
Category Managers	Category assortment, product counts, pricing and availability patterns.
Brand Managers	Brand catalogue size and average customer-interest indicators.
Pricing / Merchandising Teams	Price-segment mix and category-level price comparisons.
E-commerce Managers	Catalogue availability and product engagement overview.
Business Leadership	A concise summary of KPIs, findings, limitations and possible actions.

6. Key Business Questions

Analysis area	Business questions
Product & Brand Performance	Which categories contain the most products? Which brands have higher average loves, when minimum catalogue size is considered? Which products receive the most loves? How does out-of-stock rate vary by category?
Pricing & Positioning	How is the catalogue distributed across budget, mid-range and premium price segments? Which categories and brands have higher average prices? How do average ratings compare across price segments?
Customer Engagement	Which categories have higher average loves per product? How do brand-level loves compare with ratings? How does average loves vary across price segments? What is the rating-band distribution?

7. Functional Requirements

ID	Requirement
FR01	Load and profile the raw product catalogue.
FR02	Clean missing values, duplicates and data types using documented rules.
FR03	Create analytical features such as price segments and rating bands where appropriate.
FR04	Load the cleaned dataset into a PostgreSQL staging table.
FR05	Transform the staging data into a final reporting table and validate row and key counts.
FR06	Create SQL analysis queries and Power BI measures for agreed KPIs.
FR07	Deliver three interactive Power BI pages for product/brand, pricing and engagement analysis.
FR08	Document findings, recommendations, assumptions and limitations.

MEASUREMENT, DELIVERY & GOVERNANCE

8. Key Performance Indicators

KPI	Business meaning
Total Products / Brands / Categories	Measures catalogue scale and breadth.
Average Product Rating	Average rating across products with a recorded rating.
Average & Median Product Price	Describes typical catalogue pricing and reduces reliance on the average alone.
Premium Product Count	Counts products assigned to the premium price segment.
Out-of-Stock Rate	Share of catalogue records marked out of stock in the snapshot.
Online-Only / Exclusive Share	Share of products flagged as online-only or Sephora-exclusive.
Total Loves / Average Loves per Product	Customer-interest indicators recorded in the catalogue.
Total Reviews	Number of reviews recorded in the catalogue.

9. Technology and Delivery Approach

Technology	Role in the project
Python (Pandas, NumPy)	Data profiling, cleaning, EDA and feature engineering.
Matplotlib / Seaborn	Exploratory charts and distribution checks.
PostgreSQL	Staging, transformation, final reporting table and SQL validation.
Power BI	KPI measures, interactive visuals and business reporting.
GitHub	Version control, project documentation and portfolio presentation.

10. Scope and Limitations

In scope: catalogue structure, category and brand comparisons, price segmentation, snapshot availability, customer-interest indicators and product rating distribution.

Out of scope: actual sales, revenue, profit or margin, conversion rate, customer-level purchase behaviour, repeat purchases, historical stockout duration and sales forecasting.

Critical interpretation note: Loves and reviews are engagement indicators, not sales or orders. Out-of-stock status is a snapshot flag, not evidence of historical stockouts. The source is Sephora catalogue data; results must not be presented as actual Nykaa, Myntra or other retailer data.

11. Expected Business Outcomes and Success Criteria

The expected outcome is a reusable, validated catalogue analysis that helps business users identify assortment concentration, compare brand interest, understand price positioning and spot categories with higher snapshot unavailability. Recommendations should be proportional to the available evidence.

- All 8,494 source records are accounted for in the final reporting table, with unique product IDs validated.
- Python and SQL validation totals reconcile for key counts and selected metrics.
- Three Power BI analysis pages answer the defined business questions.
- KPI definitions, cleaning decisions, data limitations and recommendations are documented.
- Dashboard findings can be explained clearly to a non-technical business stakeholder.

Prepared for: Beauty E-commerce Analytics Portfolio **Project:** BEAUTY 360° – Beauty Catalogue Intelligence
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